



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS (FSKM)

CSF3133 WEB-BASED INTERFACE DESIGN (K4)

LAB 6

SEMESTER I 2025/2026

PREPARED BY:

CHE KU NURULHUDA BINTI CHE KU SAIFUL MUZAHAR (S75871)

LECTURER:

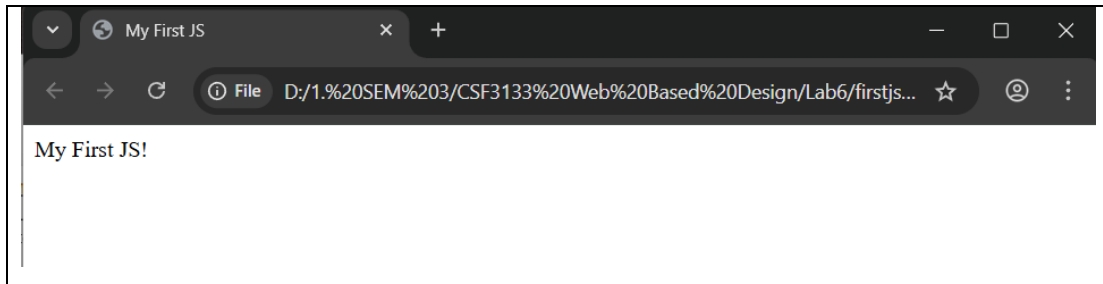
TS. DR. WAN NURAL JAWAHIR BINTI HJ WAN YUSSOF

LAB MODERATOR:

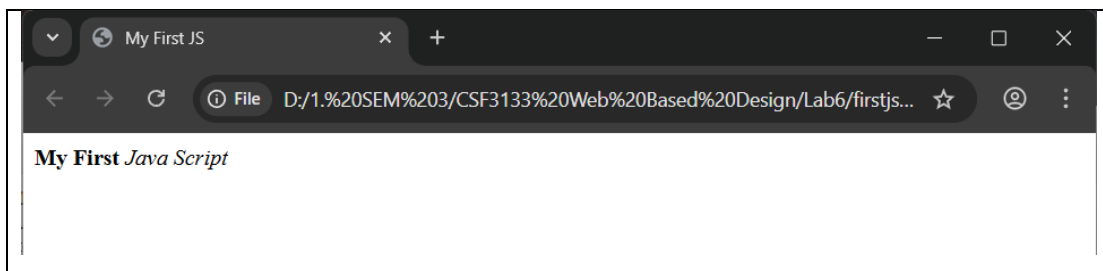
MRS. WAN FATIN FATIAH BINTI YAHYA

1. Activity 6.3 – Basic Javascript

1.1.Task 1-2

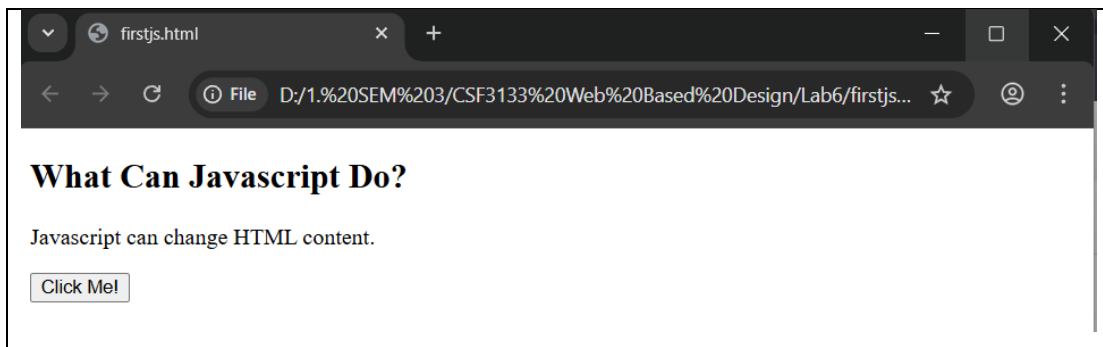


1.2.Task 3-4

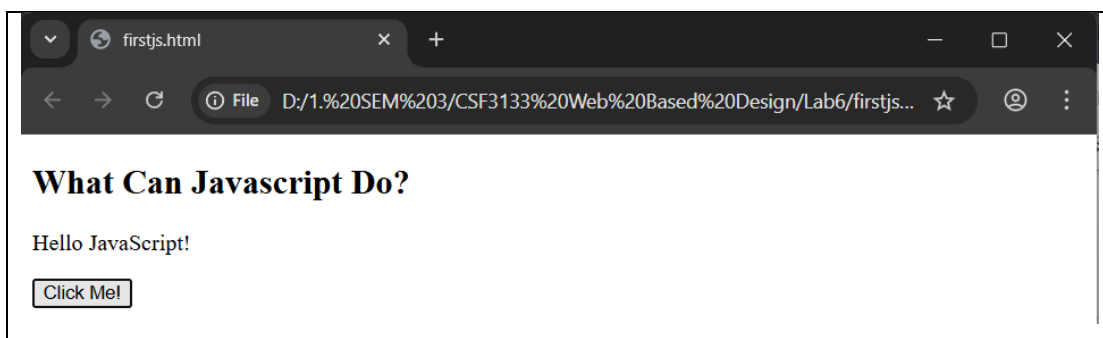


2. Activity 6.4 – getElementById()

2.1.Output before clicking the “Click Me!” button

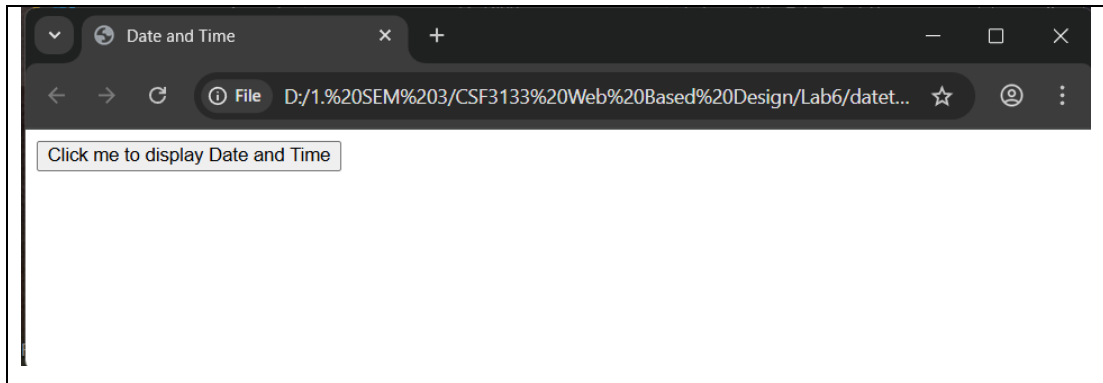


1.1.Output after clicking the “Click Me!” button

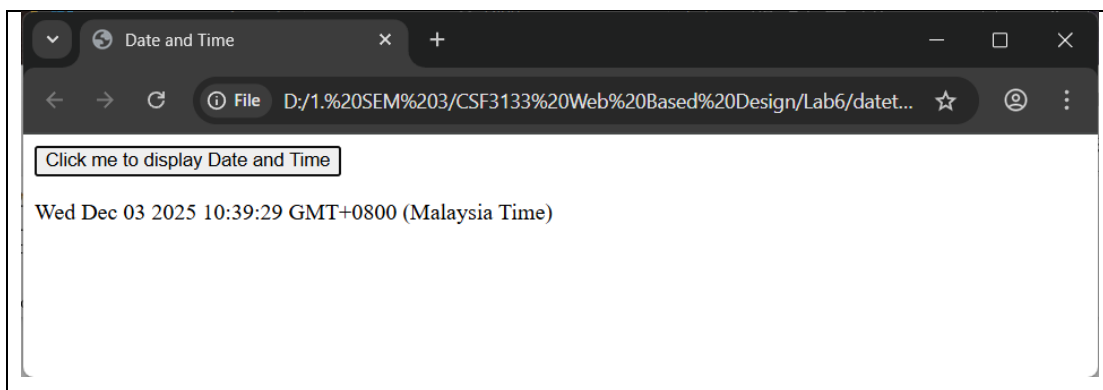


2. Activity 6.5 – Date and Time

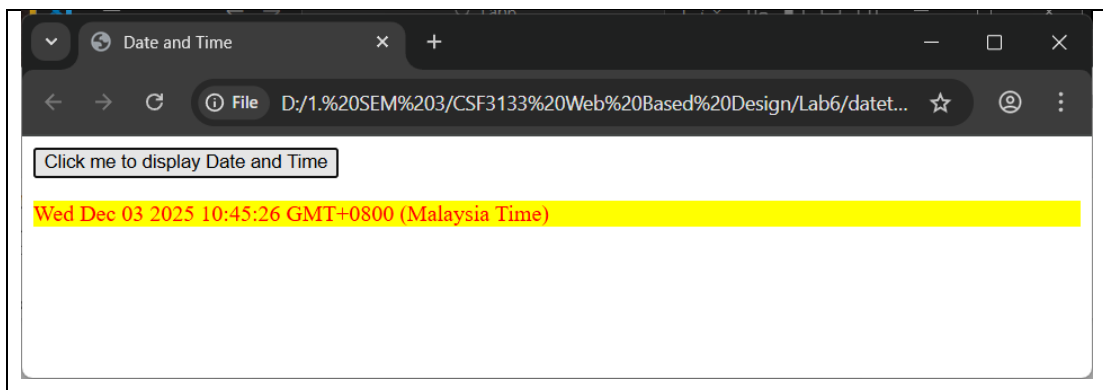
2.1.Task 1: Output before clicking the “Click me to display Date and Time” button



2.2.Task 1: Output after clicking the “Click me to display Date and Time” button

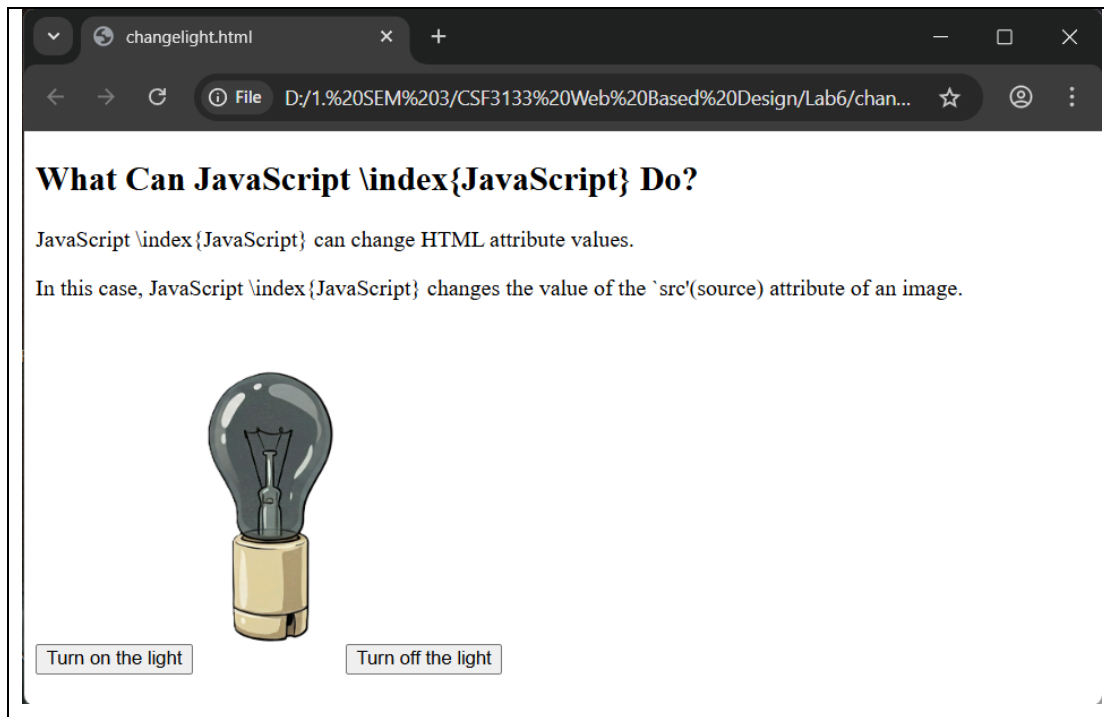


2.3.Task 2 : Output after changing the style of variable (demo)

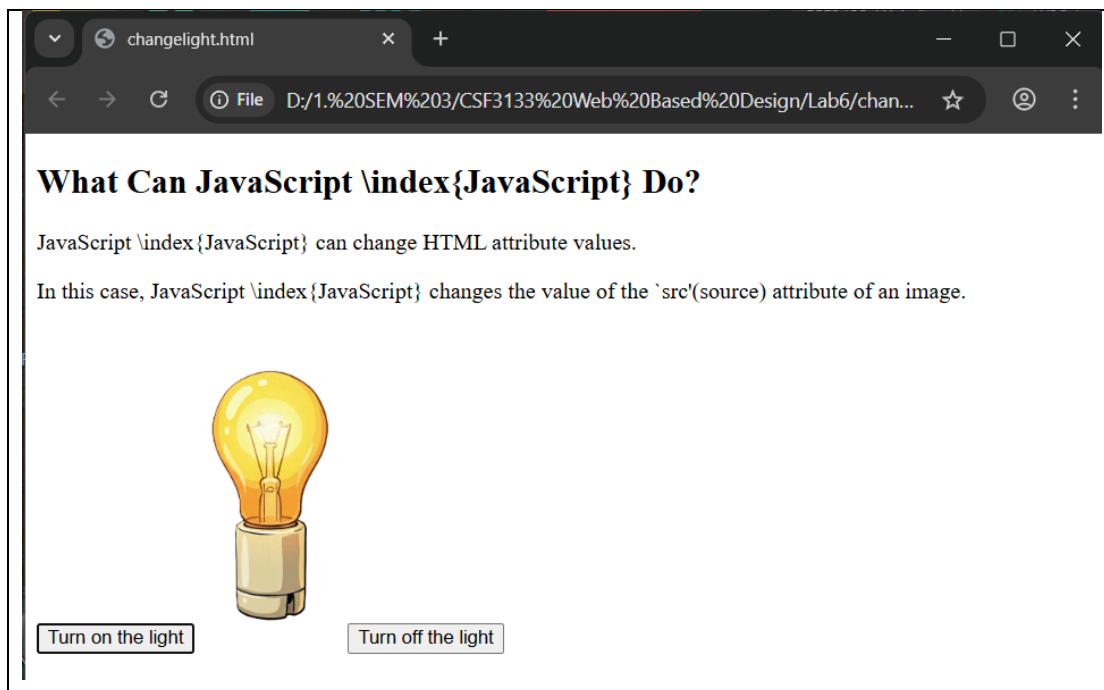


3. Activity 6.6 – Change HTML Attribute Values

3.1. Output for clicking button “Turn off the light”



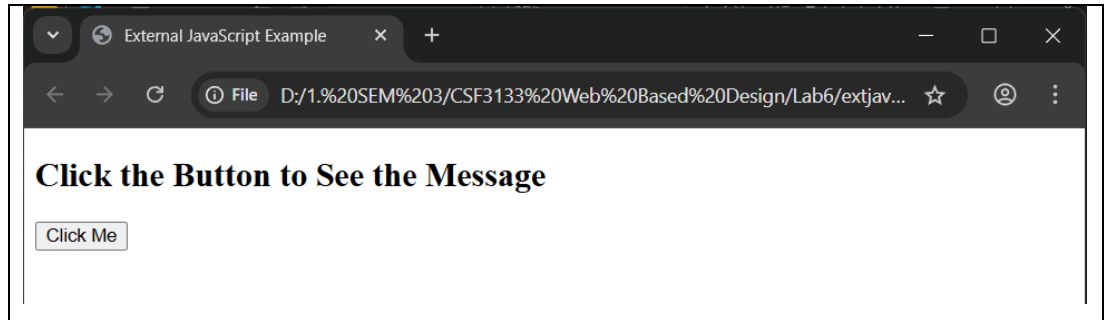
3.2. Output for clicking button “Turn on the light”



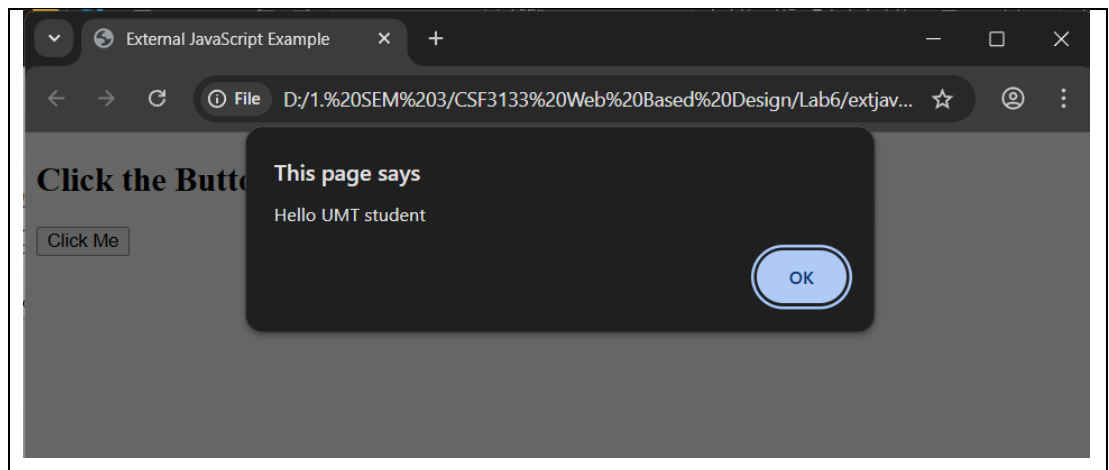
4. Activity 6.7 – Link to file Javascript (*.js)

4.1. Activity 6.7.3 : External References

4.1.1. Output before clicking the “Click Me!” button

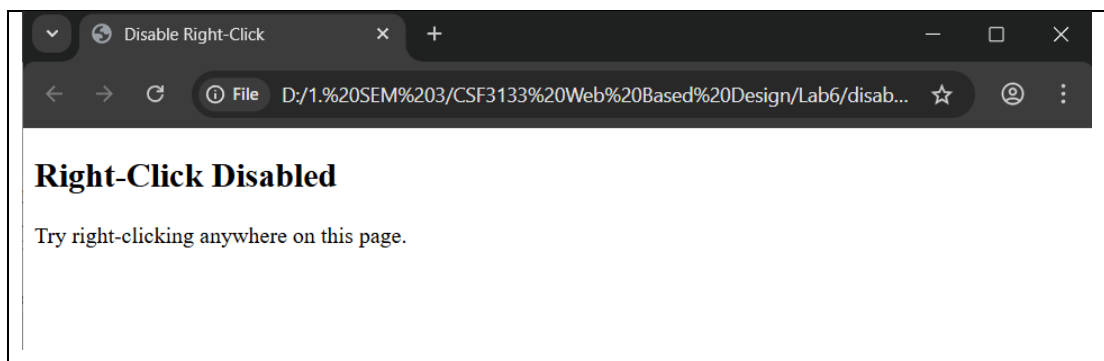


1.1.1.1. Output after clicking the “Click Me!” button

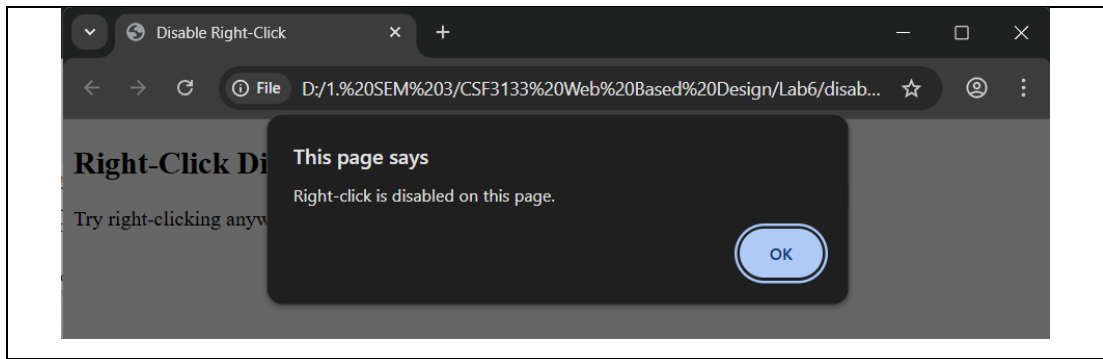


2. Activity 6.8 – Working with events

2.1. Output before right-clicking anywhere on the page

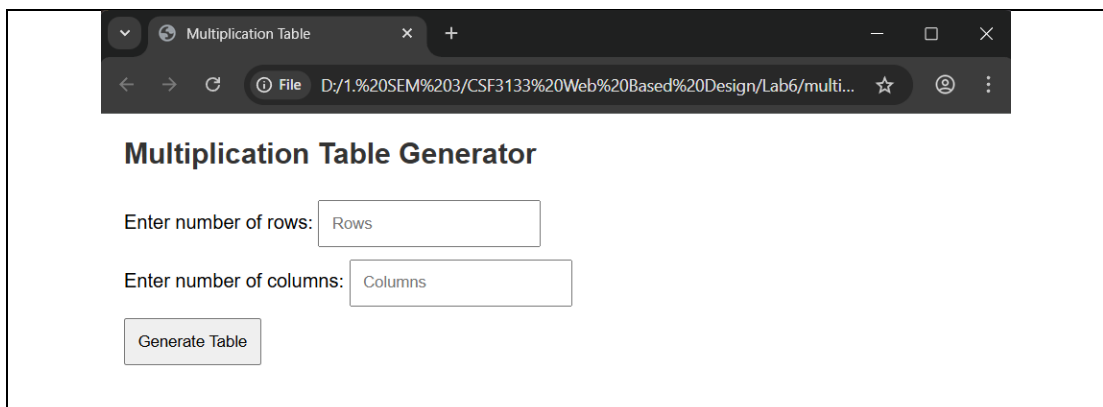


2.2. Output after right-clicking anywhere on the page



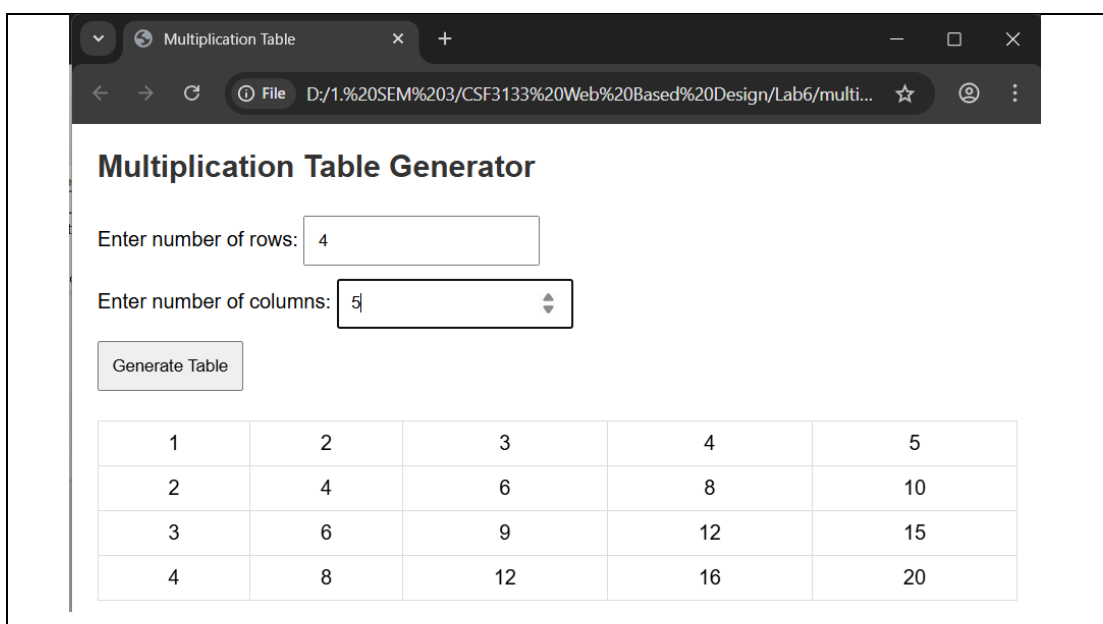
3. Activity 6.9 – Multiplication Table

3.1. Output before entering the desired number of rows and columns in the input fields.



3.2. Output after entering the desired number of rows and columns in the input fields.

Then, click the Generate Table button.

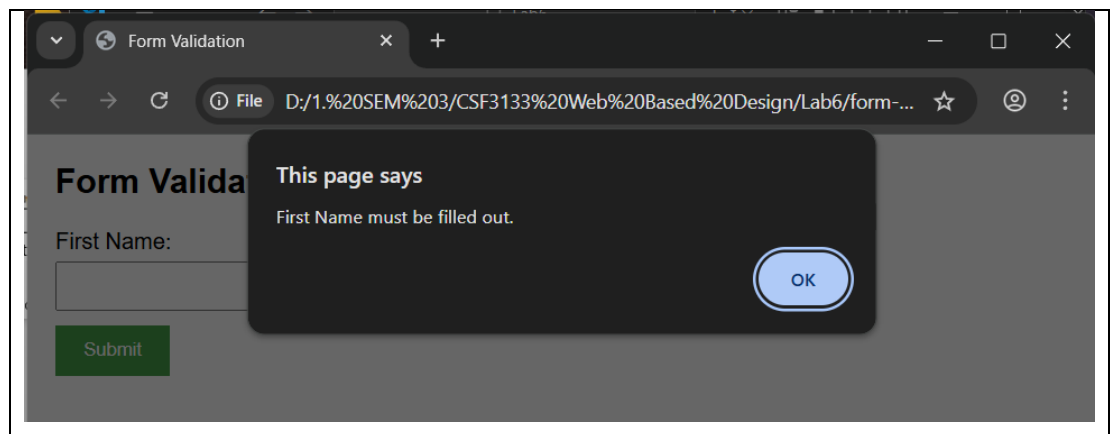


4. Activity 6.10 - Client-side validation

4.1. Activity 6.10.1 : Form validation

4.1.1. Output

4.1.2. Case 1: Leave the "First Name" field empty and click Submit. An alert should appear saying, "First Name must be filled out.".



4.1.3. Case 2: Fill in the "First Name" field and click Submit. The form should submit successfully.

4.1.3.1. Before submission

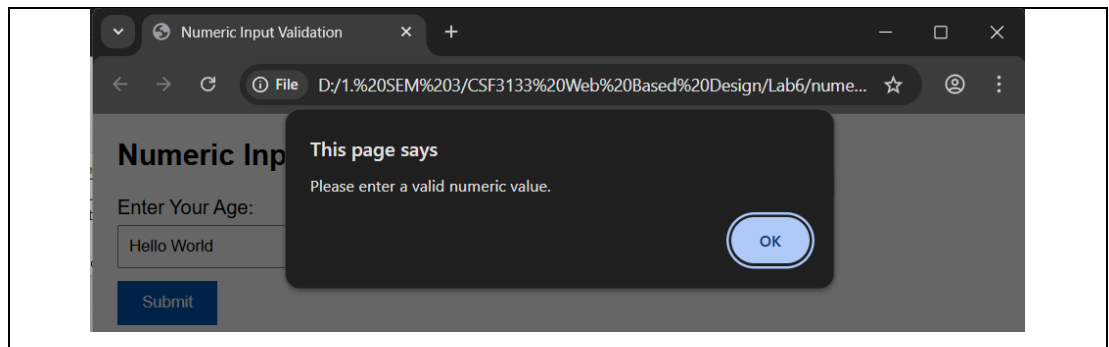
4.1.3.2. After submission

4.2. Activity 6.10.2 : Numeric Input Validation

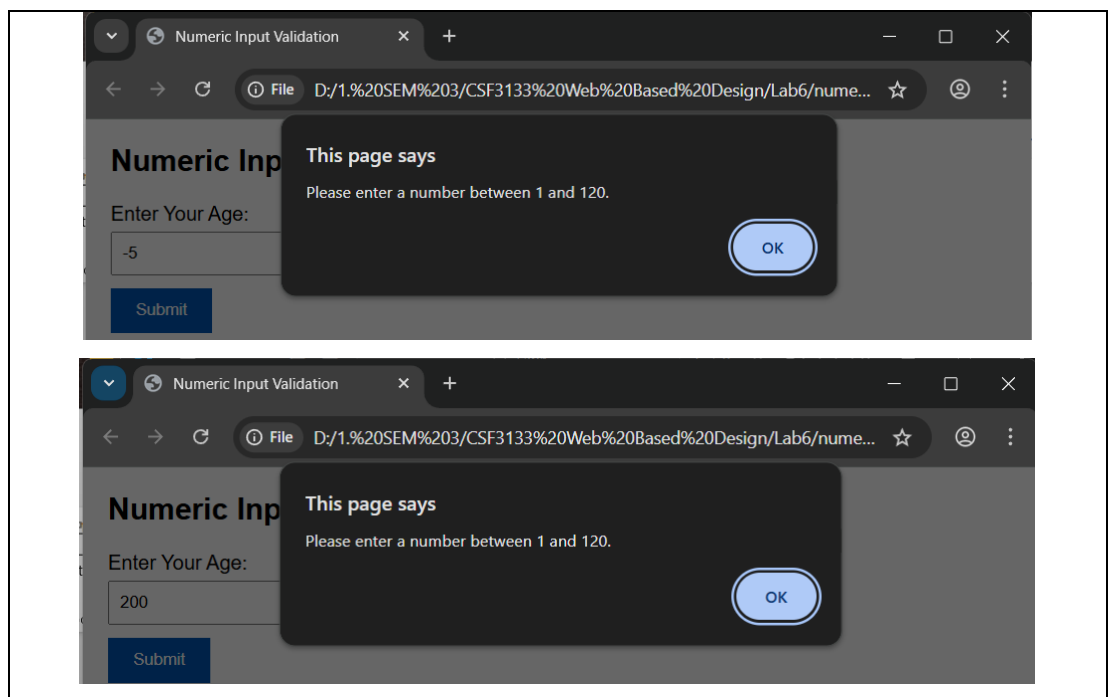
4.3. Output

4.4. Case 1: Leave the input field empty and click Submit. An alert should appear saying, "Age field cannot be empty."

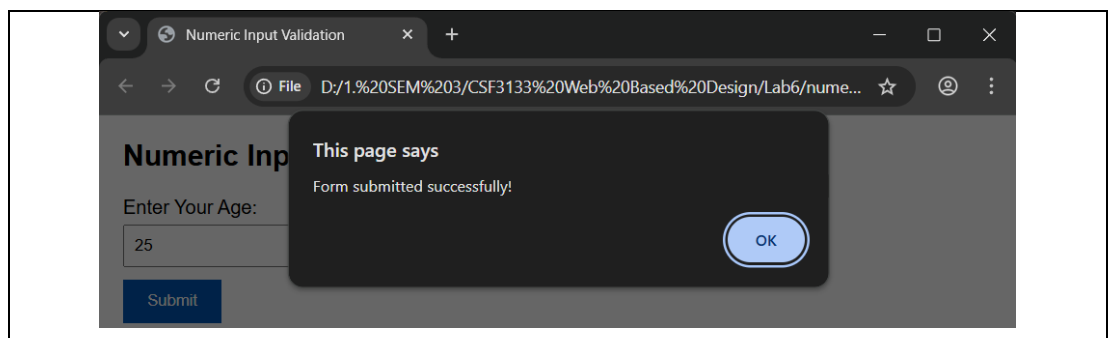
4.5. Case 2: Enter non-numeric characters and click Submit. An alert should appear saying, "Please enter a valid numeric value".



4.6. Case 3: Enter a number outside the range (-5 or 200) and click Submit. An alert should appear saying, "Please enter a number between 1 and 120.".



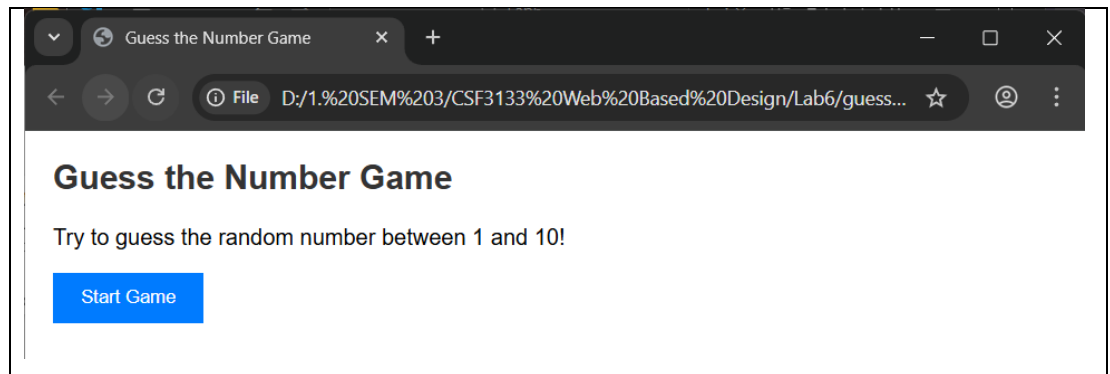
4.7. Case 4: Enter a valid number (25) and click Submit. A success message will appear.



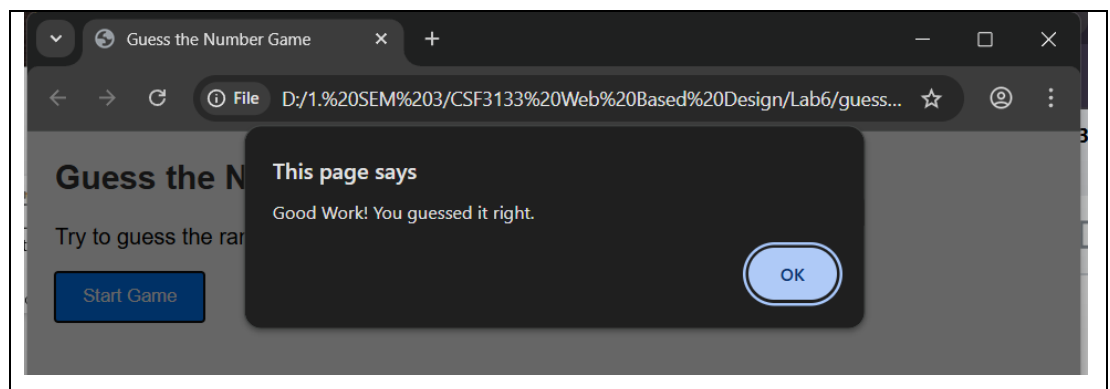
5. Activity 6.11 – Math Function

5.1. Activity 6.11.1: Guess a Number

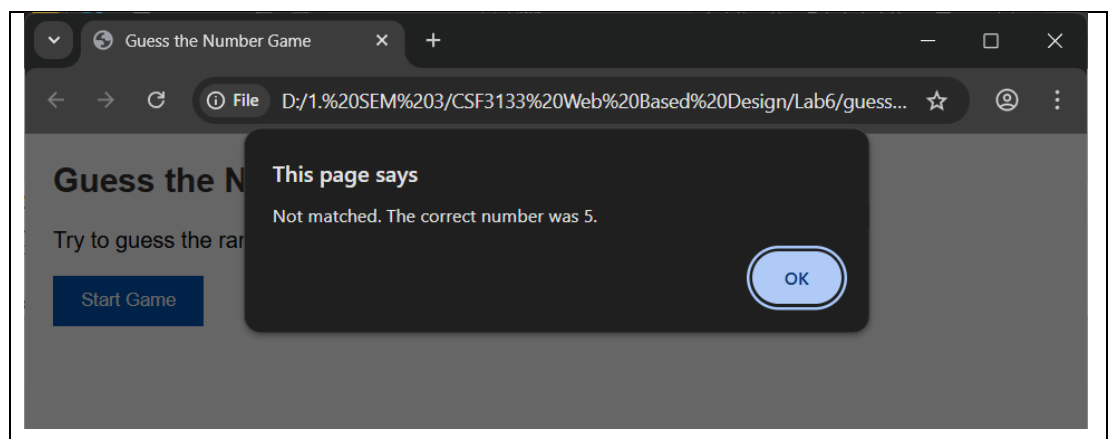
5.1.1. Output



5.1.2. Output if the guess matches the random number



5.1.3. Output if the guess is incorrect, along with the correct number



6. Activity 6.12 – Lab Exercise

6.1. Index.html

```
index.html > html > body > footer > p
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Dynamic Quiz Application</title>
7      <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10     <!-- A header that introduces the quiz.-->
11     <header>
12         <h1></h1>&#x1F3AF; Interactive Quiz Challenge</h1>
13         <p>Test your knowledge with randomized questions!</p>
14     </header>
15
16     <main>
17         <!-- A section to display the questions, answer options, and a timer. -->
18         <section id="quiz-container" class="hidden">
19             <!-- Timer display -->
20             <div id="timer-container">
21                 <span>&#x23F3; Time Left: <span id="timer">30</span>s</span>
22             </div>
23
24             <!-- Question display -->
25             <div id="question-container">
26                 <h2 id="question-number">Question 1 of 5</h2>
27                 <p id="question-text"></p>
28             </div>
29
30             <!-- Answer options display -->
31             <div id="options-container"></div>
32
33             <!-- Section to show feedback (correct/incorrect) after each answer -->
34             <div id="feedback-container" class="hidden">
35                 <p id="feedback-message"></p>
36             </div>
37
38             <!-- Button to submit answers and navigate between questions -->
39             <button id="next-btn" class="hidden">Next Question</button>
40         </section>
41
42         <!-- Start screen section -->
43         <section id="start-container">
44             <div class="start-box">
45                 <h2>Ready to Start?</h2>
46                 <p>&#x23F1; You'll have 30 seconds per question</p>
47                 <p>&#x1F4CB; Total Questions: 5</p>
48                 <button id="start-btn">Start Quiz</button>
49             </div>
50         </section>
51
52         <!-- Result screen section -->
53         <section id="result-container" class="hidden">
54             <div class="result-box">
55                 <h2>Quiz Completed!</h2>
56                 <p id="final-score"></p>
57                 <p id="performance-message"></p>
58                 <button id="restart-btn">Try Again</button>
59             </div>
60         </section>
61     </main>
62 </body>
63 </html>
```

```

60     </section>
61 </main>
62
63 <!-- Footer to display the user's score after finishing the quiz -->
64 <footer>
65     <p id="score-display">Score: <span id="current-score">0</span>
66     / <span id="total-questions">5</span></p>
67     <p>&copy; 2025 Quiz App | CHE KU NURULHUDA </p>
68 </footer>
69
70 <!-- Link to external JavaScript file -->
71 <script src="script.js"></script>
72 </body>
73 </html>|

```

6.2. style.css

```

# style.css > body
1  body {
2      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
3      background: linear-gradient(to bottom, #6500c4, #870082);
4      min-height: 100vh;
5      display: flex;
6      flex-direction: column;
7      color: #000000;
8      text-align: center;
9  }
10
11  /* Header Styles */
12  header {
13      background-color: rgba(255, 255, 255, 0.95);
14      padding: 5px;
15      box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
16  }
17
18  header h1 {
19      color: #667eea;
20      margin-bottom: 5px;
21  }
22
23  header p {
24      color: #000000;
25  }
26
27  /* Main Content */
28  main {
29      flex: 1;
30      display: flex;
31      justify-content: center;
32      align-items: center;

```

```

33     padding: 10px;
34 }
35
36 /* Quiz Container */
37 #quiz-container {
38     background: white;
39     border-radius: 30px;
40     padding: 20px;
41     width: 100%;
42     box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
43     max-width: 600px;
44 }
45
46 /* Timer */
47 #timer-container {
48     background: linear-gradient(to bottom, #ea66c0 20%, #b50000 100%);
49     color: white;
50     padding: 5px;
51     border-radius: 10px;
52     font-size: 1.7em;
53     font-weight: bold;
54 }
55
56 #question-number {
57     color: #667eea;
58     font-size: 1.2em;
59 }
60
61 #question-text {
62     font-size: 1.4em;
63     color: #2f0047;
64     line-height: 1.6;
65 }
66
67 /* Options Container */
68 #options-container {
69     display: flex;
70     flex-direction: column;
71     gap: 10px;
72     margin-bottom: 5px;
73 }
74
75 .option-btn {
76     background: linear-gradient(135deg, #e0e7ff 0%, #cfd9ff 100%);
77     border: 2px solid #667eea;
78     padding: 10px;
79     border-radius: 10px;
80     font-size: 1.1em;
81     cursor: pointer;
82     color: #333;
83 }
84
85 .option-btn:hover {
86     background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
87     color: white;
88 }
89
90 .option-btn.correct {
91     background: linear-gradient(135deg, #84fab0 0%, #8fd3f4 100%);
92     border-color: #10b981;
93     color: #065f46;
94 }

```

```

95
96 .option-btn.incorrect {
97     background: linear-gradient(135deg, #fa709a 0%, #fee140 100%);
98     border-color: #ef4444;
99     color: #7f1d1d;
100 }
101
102 .option-btn.disabled {
103     cursor: not-allowed;
104     opacity: 0.7;
105 }
106
107 /* Feedback Container */
108 #feedback-container {
109     background: #f3f4f6;
110     padding: 20px;
111     border-radius: 10px;
112     margin-bottom: 20px;
113 }
114
115 #feedback-message {
116     font-size: 1.2em;
117     font-weight: bold;
118 }
119
120 #feedback-container.correct {
121     background: #d1fae5;
122     color: #065f46;
123 }
124
125 #feedback-container.incorrect {
126     background: #fee2e2;
127     color: #7f1d1d;
128 }
129
130 /* Buttons */
131 button {
132     background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
133     color: white;
134     border: none;
135     padding: 15px 40px;
136     font-size: 1.1em;
137     border-radius: 10px;
138     cursor: pointer;
139     font-weight: bold;
140 }
141
142 button:hover {
143     box-shadow: 0 10px 20px rgba(102, 126, 234, 0.4);
144 }
145
146 /* Start Container */
147 .start-box {
148     background: white;
149     border-radius: 15px;
150     padding: 50px;
151     max-width: 500px;
152     box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
153 }
154
155 .start-box h2 {
156     color: #667eea;
157     font-size: 2em;

```

```
158     margin-bottom: 10px;
159 }
160
161 .start-box p {
162     font-size: 1.2em;
163     color: #666;
164     margin-bottom: 10px;
165 }
166
167 /* Result Container */
168 .result-box {
169     background: white;
170     border-radius: 25px;
171     padding: 50px;
172     max-width: 500px;
173     box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
174 }
175
176 .result-box h2 {
177     color: #667eea;
178     font-size: 2.2em;
179     margin-bottom: 10px;
180 }
181
182 #final-score {
183     font-size: 2.5em;
184     font-weight: bold;
185     color: #667eea;
186     margin-bottom: 10px;
187 }
188
189 #performance-message {
190     font-size: 1.3em;
191     color: #666;
192     margin-bottom: 10px;
193 }
194
195 /* Footer */
196 footer {
197     background-color: rgba(255, 255, 255, 0.95);
198     padding: 3px;
199     text-align: center;
200     box-shadow: 0 -4px 6px rgba(0, 0, 0, 0.1);
201 }
202
203 #score-display {
204     font-size: 2em;
205     font-weight: bold;
206     color: #667eea;
207     margin-bottom: 3px;
208 }
209
210 footer p:last-child {
211     color: #666;
212     font-size: 0.9em;
213 }
214
215 /* Utility Classes */
216 .hidden {
217     display: none !important;
218 }
```

6.3. script.js

```
JS script.js > questions
1 // Questions Array: Stores the questions, possible options, and the correct answer.
2 const questions = [
3   {
4     question: "Which HTML tag is used to define the largest heading?",
5     options: ["&lt;heading&gt;", "&lt;h6&gt;", "&lt;h1&gt;", "&lt;head&gt;"],
6     correct: 2 // <h1>
7   },
8   {
9     question: "Which HTML tag is used to create a hyperlink?",
10    options: ["&lt;link&gt;", "&lt;href&gt;", "&lt;a&gt;", "&lt;url&gt;"],
11    correct: 2 // <a>
12  },
13  {
14    question: "Which attribute is used to specify an image source?",
15    options: ["alt", "href", "src", "title"],
16    correct: 2 // src
17  },
18  {
19    question: "Which tag is used to insert a line break in HTML?",
20    options: ["&lt;break&gt;", "&lt;lb&gt;", "&lt;br&gt;", "&lt;newline&gt;"],
21    correct: 2 // <br>
22  },
23  {
24    question: "Which HTML element is used to define a paragraph?",
25    options: ["&lt;para&gt;", "&lt;text&gt;", "&lt;paragraph&gt;", "&lt;p&gt;"],
26    correct: 3 // <p>
27  },
28  {
29    question: "Which HTML tag is used to insert an image?",
30    options: ["&lt;image&gt;", "&lt;img&gt;", "&lt;picture&gt;", "&lt;src&gt;"],
31    correct: 1 // <img>
32  },
33  {
34    question: "Which HTML element is used to create an unordered list?",
35    options: ["&lt;li&gt;", "&lt;list&gt;", "&lt;ol&gt;", "&lt;ul&gt;"],
36    correct: 3 // <ul>
37  }
38 ];
39
40
41 // Variables to track quiz progress
42 let currentQuestion = 0;
43 let score = 0;
44 let timeLeft = 30;
45 let timer;
46 let shuffledQuestions = []; // will hold shuffled questions
47
48 document.addEventListener('DOMContentLoaded', function() {
49   const startBtn = document.getElementById('start-btn');
50   const nextBtn = document.getElementById('next-btn');
51   const restartBtn = document.getElementById('restart-btn');
52
53   startBtn.addEventListener('click', startQuiz);
54   nextBtn.addEventListener('click', nextQuestion);
55   restartBtn.addEventListener('click', restartQuiz);
56
57   // shuffleQuestions(): Randomly shuffles the questions.
58   function shuffleQuestions() {
59     const copy = [...questions];
60     copy.sort(() => Math.random() - 0.5);
61     return copy;
62   }
63 }
```

```

64 // startQuiz(): shuffles questions, shows first question, starts timer.
65 function startQuiz() {
66     document.getElementById('start-container').classList.add('hidden');
67     document.getElementById('quiz-container').classList.remove('hidden');
68
69     score = 0;
70     currentQuestion = 0;
71     document.getElementById('current-score').textContent = score;
72
73     shuffledQuestions = shuffleQuestions();
74     displayQuestion(); // NOT showQuestion()
75 }
76
77 // displayQuestion(): Displays the current question and answer options.
78 function displayQuestion() {
79     const q = shuffledQuestions[currentQuestion];
80
81     document.getElementById('question-number').innerHTML =
82         `Question ${currentQuestion + 1} of 5`;
83     document.getElementById('question-text').innerHTML = q.question;
84
85     const optionsContainer = document.getElementById('options-container');
86     optionsContainer.innerHTML = '';
87
88     for (let i = 0; i < q.options.length; i++) {
89         const button = document.createElement('button');
90         button.innerHTML = q.options[i];
91         button.classList.add('option-btn');
92         button.onclick = function () {
93             checkAnswer(i);
94         };
95         optionsContainer.appendChild(button);
96     }
97
98     document.getElementById('feedback-container').classList.add('hidden');
99     document.getElementById('next-btn').classList.add('hidden');
100
101     startTimer();
102 }
103
104 // startTimer(): Sets a countdown timer for answering each question.
105 function startTimer() {
106     timeLeft = 30;
107     document.getElementById('timer').textContent = timeLeft;
108     clearInterval(timer);
109
110     timer = setInterval(function () {
111         timeLeft--;
112         document.getElementById('timer').textContent = timeLeft;
113
114         if (timeLeft <= 0) {
115             clearInterval(timer);
116             timeUp();
117         }
118     }, 1000);
119 }
120
121 function timeUp() {
122     const buttons = document.querySelectorAll('.option-btn');
123     for (let i = 0; i < buttons.length; i++) {
124         buttons[i].disabled = true;

```

```

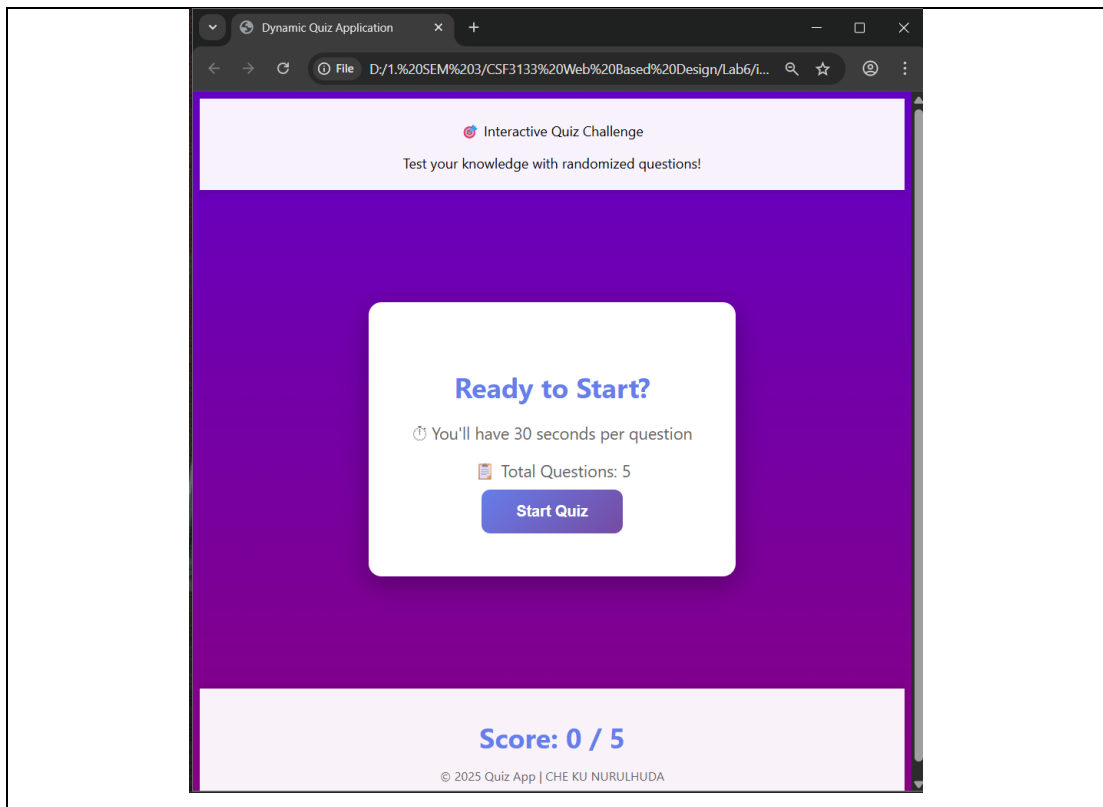
125     }
126     showFeedback(false, "⏱ Time's up!");
127     document.getElementById('next-btn').classList.remove('hidden');
128 }
129
130 // checkAnswer():
131 // Checks whether the selected answer is correct and updates the score.
132 function checkAnswer(selectedAnswer) {
133     clearInterval(timer);
134
135     const q = shuffledQuestions[currentQuestion];
136     const buttons = document.querySelectorAll('.option-btn');
137
138     for (let i = 0; i < buttons.length; i++) {
139         buttons[i].disabled = true;
140     }
141
142     if (selectedAnswer === q.correct) {
143         score++;
144         document.getElementById('current-score').textContent = score;
145         buttons[selectedAnswer].classList.add('correct');
146         showFeedback(true, "✅ Correct! Well done!");
147     } else {
148         buttons[selectedAnswer].classList.add('incorrect');
149         buttons[q.correct].classList.add('correct');
150         showFeedback(false, "❌ Incorrect. The correct answer is:
151             ${q.options[q.correct]}");
152     }
153
154     document.getElementById('next-btn').classList.remove('hidden');
155 }
156
157 function showFeedback(isCorrect, message) {
158     const feedbackBox = document.getElementById('feedback-container');
159     feedbackBox.classList.remove('hidden');
160     document.getElementById('feedback-message').innerHTML = message;
161
162     if (isCorrect) {
163         feedbackBox.classList.add('correct');
164         feedbackBox.classList.remove('incorrect');
165     } else {
166         feedbackBox.classList.add('incorrect');
167         feedbackBox.classList.remove('correct');
168     }
169 }
170
171 // nextQuestion(): Moves to the next question,
172 // resetting the timer and updating the score.
173 function nextQuestion() {
174     currentQuestion++;
175
176     if (currentQuestion < 5) {
177         displayQuestion();
178     } else {
179         showResults();
180     }
181 }
182
183 function showResults() {

```

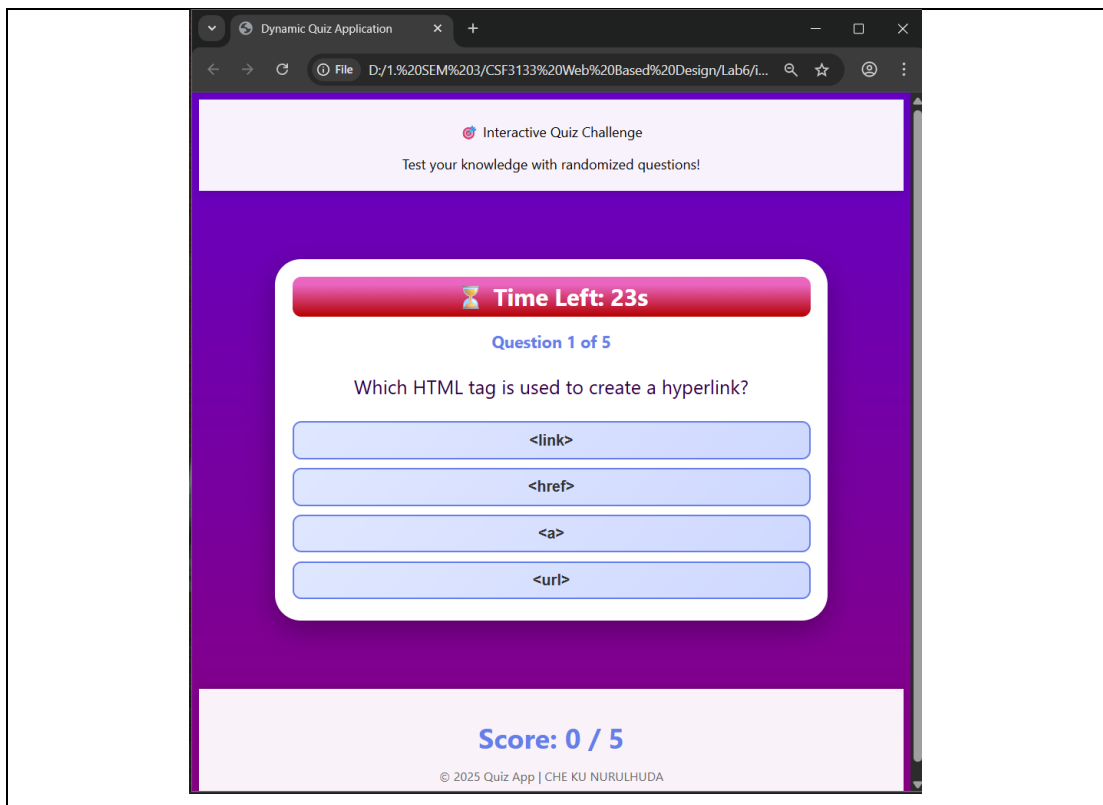
```
184     clearInterval(timer);
185     document.getElementById('quiz-container').classList.add('hidden');
186     document.getElementById('result-container').classList.remove('hidden');
187
188     document.getElementById('final-score').textContent = `${score} / 5`;
189
190     let message;
191     if (score === 5) {
192         message = "🏆 Perfect Score! You're a genius!";
193     } else if (score >= 4) {
194         message = "👏 Excellent work! Keep it up!";
195     } else if (score >= 3) {
196         message = "👍 Good job! Room for improvement!";
197     } else if (score >= 2) {
198         message = "👎 Not bad, but try studying more!";
199     } else {
200         message = "📚 Keep practicing, you'll get better!";
201     }
202
203     document.getElementById('performance-message').innerHTML = message;
204 }
205
206 function restartQuiz() {
207     clearInterval(timer);
208     document.getElementById('result-container').classList.add('hidden');
209     document.getElementById('start-container').classList.remove('hidden');
210
211     currentQuestion = 0;
212     score = 0;
213     document.getElementById('current-score').textContent = score;
214 }
215 };
```

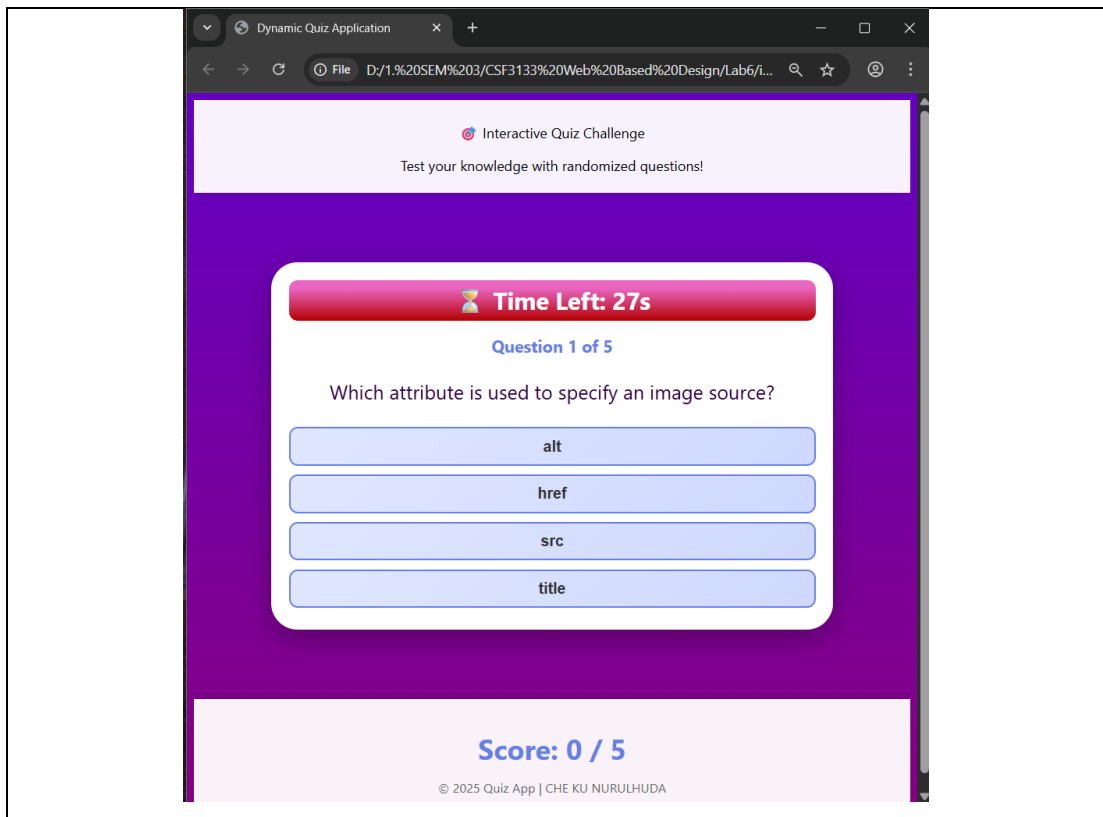
6.4. Output

6.4.1. Quiz Start Section

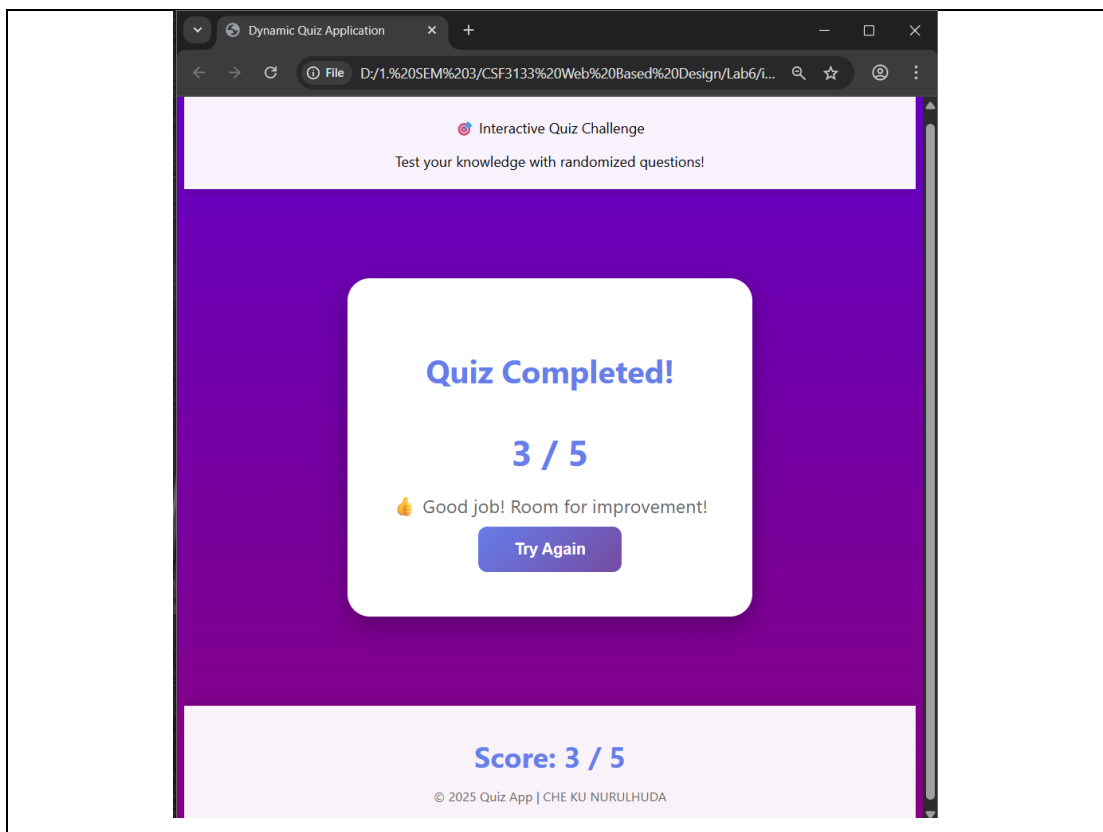


6.4.2. Quiz Questions and Options Section (Randomly shuffles question)

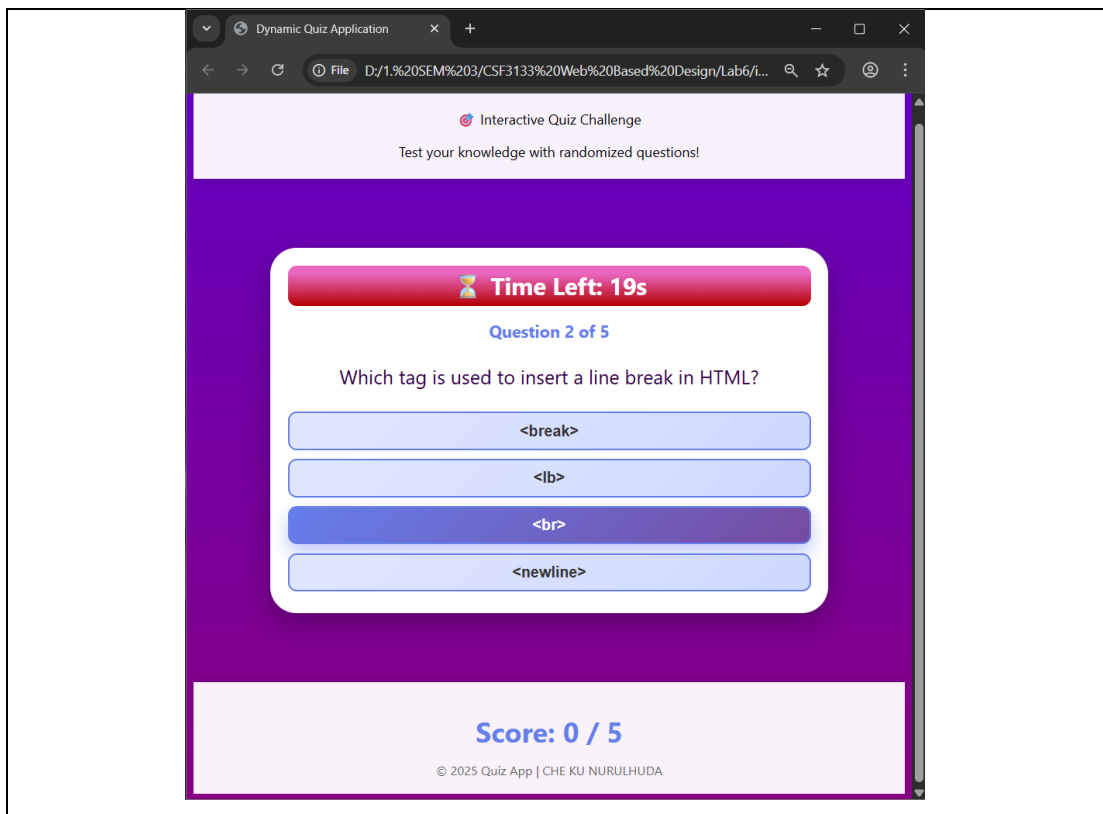




6.4.3. Quiz Result Section



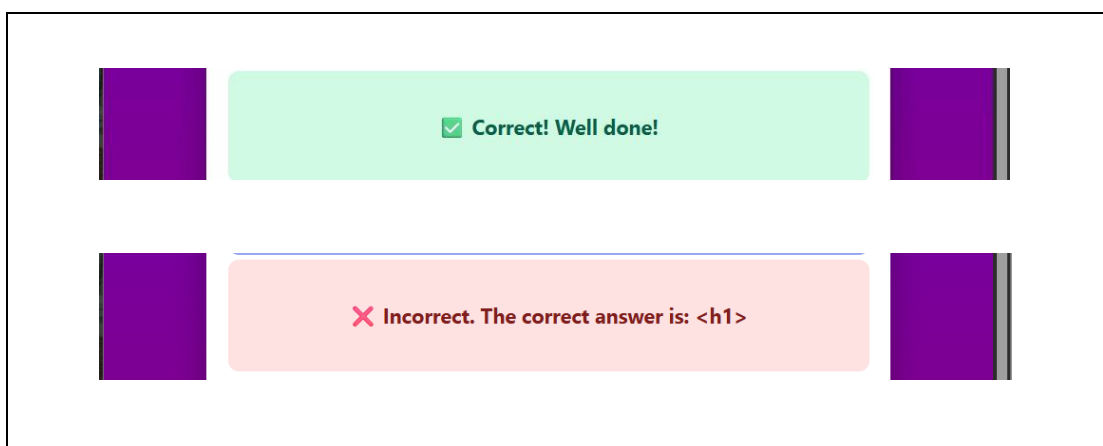
6.4.4. Hover Over Button



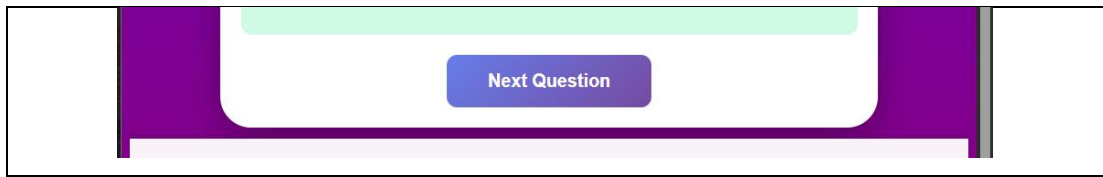
6.4.5. Timer Display



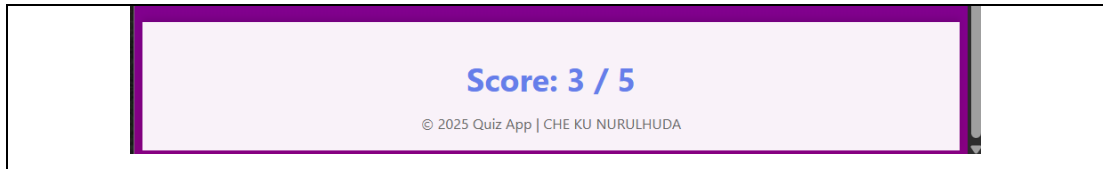
6.4.6. Feedback (correct/incorrect) after each answer



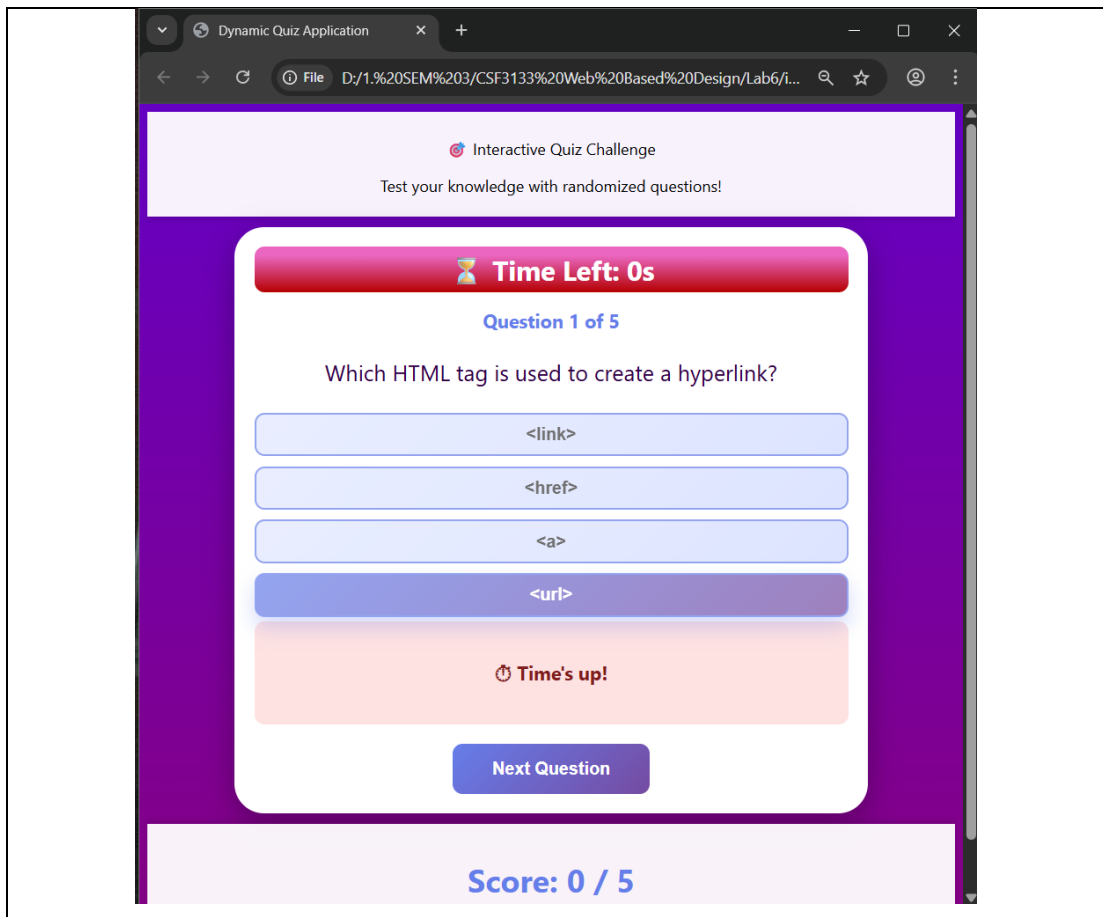
6.4.7. Submit Answers Button to Navigate Between Questions



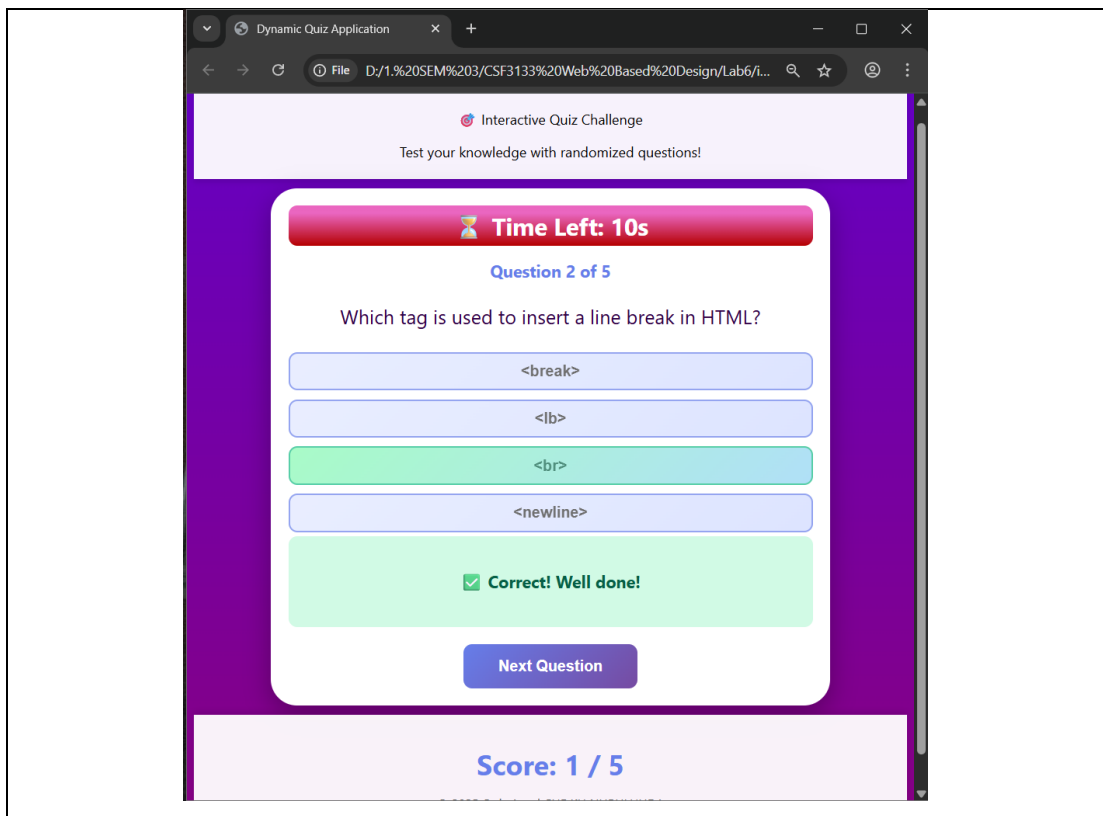
6.4.8. Footer Displaying the User's Finished Score



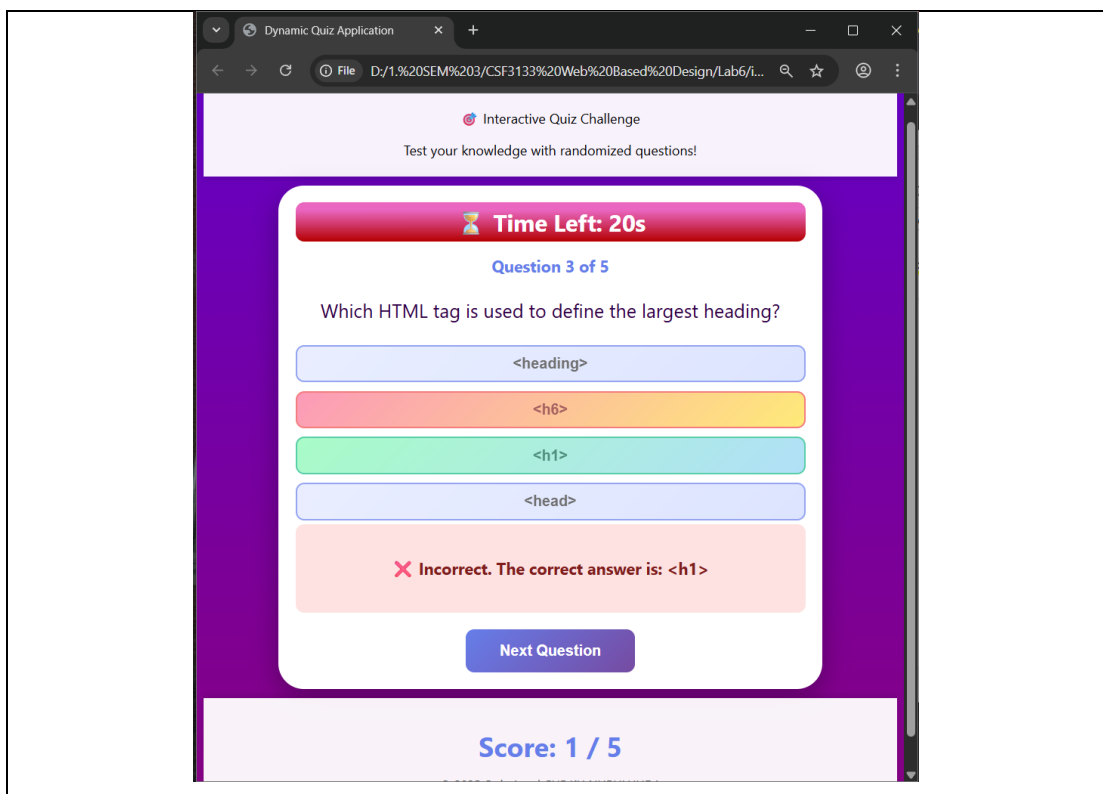
6.4.9. When the timer is up, user cannot click the answer option button.



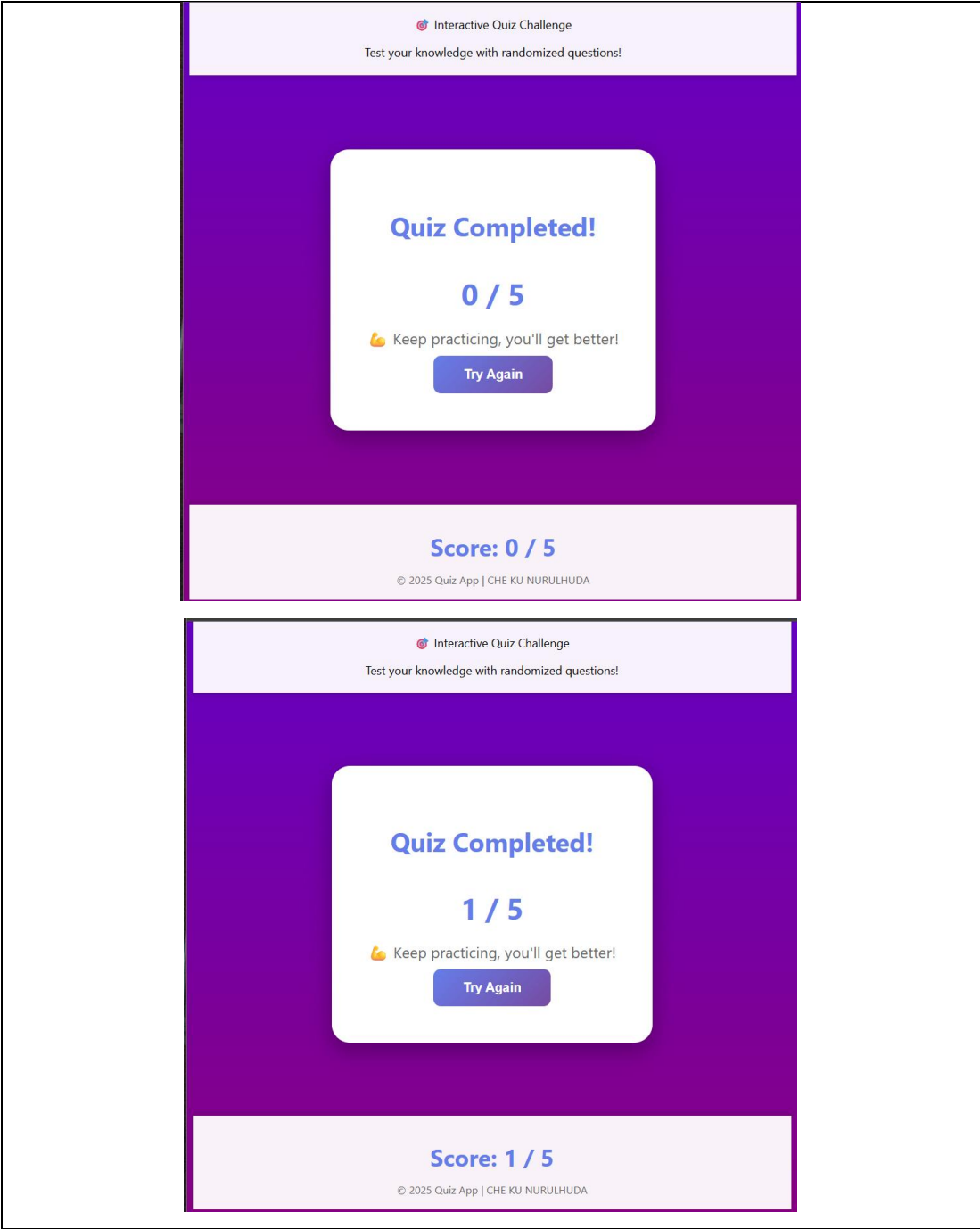
6.4.10. When user clicked the right answer



6.4.11. When user clicked the wrong answer



6.4.12. Feedback message to user for each score (1-5)



Quiz Completed!

2 / 5

Not bad, but try studying more!

Try Again

Score: 2 / 5

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Quiz Completed!

3 / 5

Good job! Room for improvement!

Try Again

Score: 3 / 5

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Quiz Completed!

4 / 5

🌟 Excellent work! Keep it up!

Try Again

Score: 4 / 5

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Quiz Completed!

5 / 5

🏆 Perfect Score! You're a genius!

Try Again

Score: 5 / 5

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